# Hadi Mohammadi

Senior AI & Data Science Expert · PhD in Explainable NLP

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## Profile

Researcher in explainable NLP (PhD, Utrecht University) and Senior AI & Data Science Expert at AcademicTransfer. I build production LLM systems for ranking, matching, and evaluation, and I study how to make them transparent, reliable, and culturally fair.

Track record: six doctoral research projects, five competitively funded projects (over €40K) led end to end, and production AI used by 22 Dutch research universities and university medical centres. I release code and data with my work.

## Professional Experience

**AcademicTransfer**, Senior AI & Data Science Expert - R&D Specialist — Utrecht, The Netherlands
Dec 2025 – present

- Lead the 2026 WBSO R&D programme as R&D Specialist (since Mar 2026), two projects approved for Mar–Dec 2026: an MCP-based academic query engine and an explainable CV matching engine with recruiters in the loop.
- Technical lead of the AI intelligence layer and applicant-tracking system used by 22 Dutch research universities and university medical centres.
- Own the AI stack: CV creation and revision, semantic search, talent-pool matching, and best-candidate identification.
- Lead a national recognition-and-rewards policy analysis with Dr. Sanli Faez (Universiteiten van Nederland) and university policy makers.
- Building an e-learning programme for PhD candidates and researchers with the University of Amsterdam.

**AcademicTransfer**, AI & Data Science Expert — Utrecht, The Netherlands
Jan 2024 – Dec 2025

- Cut HR hiring time by 20% with the CV Priority Sorter (active learning + LLMs), built with Utrecht University, TU Delft, and Maastricht University.
- Raised candidate–job matching accuracy by 17% and halved manual effort with generative pipelines for CV insight extraction and de-identification.
- Shipped transformer models on Azure ML with MLflow, Docker, and CI/CD; evaluated ranking with A/B tests, uplift analysis, and bandit simulations.

**Utrecht University, School of Economics**, Researcher & NLP Developer, FIRMBACKBONE — Utrecht, The Netherlands
Feb 2024 – Mar 2025

- Built the fbb-sustainability-analysis-cli tool: keyword extraction and sustainability scoring of company websites.

**SnowaTec (Innovation Center), Entekhab Electronic**, Senior Data Scientist, Customer Behaviour — Tehran, Iran
Dec 2021 – Jan 2023

- Modernised the enterprise data warehouse and ETL, improving data accessibility by 16%.
- Defined company KPIs and a customer-satisfaction index (+2% satisfaction); built Power BI dashboards used at every management level.
- Deployed ML customer segmentation, lifting marketing impact by 7.5%.

**Bdood.bikes (shared bicycle system)**, Head of Data Science & Business Intelligence Team — Tehran, Iran
Apr 2020 – Oct 2020

- Led the data science and BI team and its analytics roadmap.
- Launched dynamic pricing and ML usage prediction, raising revenue by 5%.
- Cut maintenance costs by 8% and relocation costs by 3% with fleet intelligence; raised station utilisation by 11%.
- Improved customer retention by 4.5% through user-behaviour analyses.

**Bdood.bikes (shared bicycle system)**, Data Scientist — Tehran, Iran
July 2019 – Apr 2020

- Reduced downtime by 3.8% with predictive maintenance and battery-health algorithms.
- Improved rider safety by 3% with GIS models identifying high-risk riders.
- Delivered marketing analytics behind a 6% rise in revenue and engagement.

## Education

**PhD** **Utrecht University**, Explainable Natural Language Processing for Social Sciences — Utrecht, The Netherlands
Feb 2023 – Oct 2025

- Defended 9 September 2026.
- Thesis: *Let Me Explain! Explainable NLP for Understanding Large Language Models*.
- Department of Methodology & Statistics; member of the NLTP group and the Human Data Science Lab.
- Supervisors: Dr. Robert A. Bagheri, Prof. Dr. Daniel Oberski, Dr. Anastasia Giachanou.

**MSc** **University of Tehran**, Industrial Engineering, Macro Systems — Tehran, Iran
Sept 2018 – Apr 2021

- GPA 18.96/20 (4/4); ranked 2nd among 50+ graduate students.
- Thesis: dynamic pricing with demand learning via reinforcement learning under non-stationary customer behaviour.
- Cognitive science research on customer behaviour with the Institute for Cognitive Science Studies (ICSS).

**BSc** **Isfahan University of Technology, Golpayegan College of Engineering**, Mechanical Engineering — Isfahan, Iran
Sept 2011 – July 2016

## Skills

**Research:** Explainable NLP, LLM evaluation and LLM-as-judge, fairness and cultural alignment, annotation reliability, human–AI collaboration.

**ML & Statistics:** Bayesian statistics, reinforcement learning, multi-armed bandits (Thompson sampling), transformers, topic modelling, A/B testing and uplift analysis, geospatial analysis, high-performance computing.

**Engineering:** Python (advanced), R, SQL, Bash; PyTorch, TensorFlow, Hugging Face, scikit-learn; RAG, agents, Model Context Protocol, vector databases; Azure ML, MLflow, Docker, CI/CD; Power BI; LaTeX.

**Languages:** Farsi (native), English (fluent), Dutch (A2, improving).

## Publications

**trajectory-judge: What Outcome-Only LLM Judges Miss on Agent Trajectories** — 2026
***Mohammadi, H.***
arxiv.org/abs/2609.00038 (NeurIPS 2026 Workshop "Who Verifies the Agents?" (accepted))

**Reasoning–Answer Entailment Under GRPO and DPO Fine-Tuning** — 2026
***Mohammadi, H.***, Giachanou, A.
mohammadi.cv/publications/grpo-dpo-cot (INLG 2026 (ACL), Utrecht, long paper, pp. 718–730)

**EvalMORAAL: Interpretable Chain-of-Thought and LLM-as-Judge Evaluation for Moral Alignment in Large Language Models** — 2026
***Mohammadi, H.***, Giachanou, A., Bagheri, R. A.
mohammadi.cv/publications/evalmoraal (*SEM 2026 (ACL), pp. 497–515)

**Exploring Cultural Variations in Moral Judgments with Large Language Models** 2026
***Mohammadi, H.***, Bagheri, R. A.
mohammadi.cv/publications/cultural-moral-judgments (Computational Linguistics in the Netherlands Journal, Vol. 15, pp. 59–77)

**Assessing the Reliability of LLMs Annotations in the Context of Demographic Bias and Model Explanation** 2025
***Mohammadi, H.***†, Shahedi, T.†, Mosteiro, P., Poesio, M., Bagheri, R. A., Giachanou, A.
mohammadi.cv/publications/llm-annotation-reliability (GeBNLP Workshop, ACL 2025, pp. 92–104 († equal contribution))

**Do Large Language Models Understand Morality Across Cultures?** 2025
***Mohammadi, H.***, Meijer, Y. F. S. S., Papadopoulou, E., Bagheri, R. A.
mohammadi.cv/publications/morality-across-cultures (2nd LUHME Workshop, ECAI 2025, pp. 30–39)

**Explainability-Based Token Replacement on LLM-Generated Text** 2025
***Mohammadi, H.***, Giachanou, A., Oberski, D. L., Bagheri, R. A.
mohammadi.cv/publications/ai-text-undetectability (arXiv preprint)

**Explainability in Practice: A Survey of Explainable NLP Across Various Domains** 2025
***Mohammadi, H.***, Bagheri, R. A., Giachanou, A., Oberski, D. L.
mohammadi.cv/publications/explainability-survey (arXiv preprint; under review at the Journal of Information Science)

**Measuring Firm Characteristics with Natural Language Processing (NLP) Models: An Error-Mitigating Framework with Tutorial Guidance** 2025
Oberski, D. L., Domenech, J., Garcia-Bernardo, J., Gerbrands, P., Hassink, W., ***Mohammadi, H.***, Mustica, P., Schilpzand, R., van Witteloostuijn, A.
(Strategic Organization (under revision))

**Novel Approaches in Financial Fraud Detection: Hybrid Machine Learning and Uncertainty-Based Deep Learning** 2024
***Mohammadi, H.***, Rahmati, M., Shahedi, T.
(BNAIC/BeNeLearn 2024, Utrecht, The Netherlands)

**A Transparent Pipeline for Identifying Sexism in Social Media: Combining Explainability with Model Prediction** 2024
***Mohammadi, H.***, Giachanou, A., Bagheri, R. A.
mohammadi.cv/publications/sexism-detection (Applied Sciences, 14(19), 8620)

**The CLIN33 Shared Task on the Detection of Text Generated by Large Language Models** 2024
Fivez, P., Daelemans, W., Van de Cruys, T., Kashnitsky, Y., Chamezopoulos, S., ***Mohammadi, H.***, Giachanou, A., Bagheri, R. A., Poelman, W., Vladika, J., Ploeger, E., Bjerva, J., Matthes, F., van Halteren, H.
clinjournal.org/clinj/article/view/182 (Computational Linguistics in the Netherlands Journal, 13, 233–259)

**Towards Robust Online Sexism Detection: A Multi-Model Approach with BERT, XLM-RoBERTa, and DistilBERT for EXIST 2023 Tasks** 2023
***Mohammadi, H.***, Giachanou, A., Bagheri, R. A.
ceur-ws.org/Vol-3497/paper-085.pdf (Working Notes of CLEF 2023, CEUR-WS Vol-3497, pp. 1000–1011)

**Statistical Reinforcement Learning with Application in Dynamic Pricing Problem: Learning from Customer Interaction based on Bayesian Approach** 2022
***Mohammadi, H.***
(Tehran: Arshadan Publication (book, in Persian))

**Home Health Care Problem with Synchronization Visits and Considering Samples Transferring Time: A Case Study in Tehran, Iran** 2022
Mirabnejad, M., ***Mohammadi, H.***, Mirzabaghi, M., Aghsami, A., Jolai, F., Yazdani, M.
10.3390/ijerph192215036 (International Journal of Environmental Research and Public Health, 19(22), 15036)

## Grants & Awards

- **Best Oral Presentation**, IOPS Winter Conference 2025 — cultural moral alignment in LLMs (Dec 2025).
- **Applied Data Science Grant, €5K** — CV Priority Sorter for fair prioritization; with R. A. Bagheri, G. Krempl, J. Sparla (Dec 2025).
- **Applied Data Science Grant, €5K** — rule extraction from human evaluation with LLMs and RL; with A. Giachanou, S. Wang (Jun 2025).
- **CUCo Spark Grant, €9K** — ReDOSE: circular pharmaceutical industry using AI; with WUR, TU/e, and UMCU collaborators (Jul 2025).
- **ENFIELD grant, €14.4K** — European Lighthouse to Manifest Trustworthy and Green AI: user perspectives on explainable AI (Dec 2024).
- **Applied Data Science Grant, €7.5K** — reliability of annotations under model predictions and explanations; with P. Mosteiro Romero, A. Giachanou, M. Poesio (Dec 2023).
- **Admission scholarship**, Explainable AI Summer School, TU Delft (Aug 2022).
- **Ranked 2nd** among 50+ Industrial Engineering graduate students, University of Tehran (2021).
- **Best Paper Award**, 2nd National Conference of the Iranian System Dynamics Society (2019).
- **Ranked 35th** of 10,000+ in the national Iranian MSc entrance exam, Industrial Engineering (2018).

## Teaching & Supervision

- **MSc thesis co-supervision, Applied Data Science, Utrecht University** — 2024: M. Meijer, E. Papadopoulou, L. Santos Moitinho de Almeida (with R. A. Bagheri); 2025: H. Brouwer, T. Kozák, M. Jouhar (with A. Giachanou); 2026: A. Vos, G. Verestiuc, T. Rietman (with R. A. Bagheri).
- **Practical Assistant**, Transformers Workshop, UMC Utrecht — Python sessions with Dr. Robert A. Bagheri and Prof. Dr. Albert Gatt (Dec 2023).
- **Computer Lab Assistant**, Data Wrangling and Data Analysis, MSc Applied Data Science, Utrecht University (Sep–Oct 2023).
- **Lead Teaching Assistant**, Deep Learning Summer School, Neuromatch Academy — international cohort, remote (Aug–Sep 2021).

## Leadership & Service

- **Co-founder, NLP@U Young Scholar Network** — community of PhD candidates, postdocs, and junior NLP researchers across Utrecht University and UMC Utrecht (Nov 2024).
- **Communications Chair, BNAIC/BeNeLearn 2024** — Joint International Scientific Conferences on AI and Machine Learning, Utrecht University.
- **LERU Doctoral Summer School representative** for the Faculty of Social and Behavioural Sciences, Utrecht University (2025).

## Professional Development

**Generative Modeling Summer School, TU Eindhoven** June 2024

**Oxford Machine Learning Summer School (OxML 2023) — ML x Finance & NLP** May – July 2023

**Explainable AI Summer School, TU Delft — post-hoc interpretability** August 2022

**Executive Master in Finance, Sharif University of Technology** Nov 2021 – Nov 2022

## References

- Dr. Robert A. Bagheri — Associate Professor, Methodology & Statistics, Utrecht University — a.bagheri@uu.nl
- Dr. Anastasia Giachanou — Assistant Professor, Methodology & Statistics, Utrecht University — a.giachanou@uu.nl
- Prof. Dr. Daniel Oberski — Professor of Social Data Science, Utrecht University; Scientific Director, ODISSEI — d.l.oberski@uu.nl